

CHEMISTRY (043)

Reconstructed Question Paper — 2025

Reconstructed from the submitted answer booklet. Chemical equations and numerical relationships visible in the working have been retained; obscured multiple-choice stems are not invented.

Section A — Multiple Choice Questions

Q.1–Q.16 Choose the correct answer. The answer booklet indicates questions covering:

1. bicarbonate/organic functional-group tests,
2. activation energy,
3. haloalkanes such as chlorobutane,
4. reaction with phosphorus halides,
5. amines and their nomenclature,
6. nitro compounds,
7. order of a reaction,
8. coordination compounds,
9. assertion–reason questions.

Section B — Solutions and Electrochemistry

Q.17 State Raoult's law for an ideal solution and explain the meaning of the partial vapour pressures.

Q.18 An equimolar solution of two volatile liquids X and Y is given. Using Raoult's law, determine the total vapour pressure/partial pressures from the supplied vapour-pressure data.

Q.19 Explain why the molar conductivity of a strong electrolyte such as KCl increases on dilution, whereas its conductivity decreases.

Q.20 Explain the meaning of order and molecularity. Why does molecularity have meaning only for an elementary reaction?

Section C — Chemical Kinetics and Organic Chemistry

Q.21 Two reactants X and Y are each first order with respect to themselves. Write the rate law and determine how the rate changes when the relevant concentrations are changed as indicated.

Q.22 Explain the mechanism of the acid-catalysed dehydration of an alcohol using concentrated H_2SO_4 at elevated temperature. Show:

- (i) protonation of the alcohol,
- (ii) formation of the carbocation/intermediate,
- (iii) deprotonation and alkene formation.

Q.23 Explain why carboxylic acids are stronger acids than phenols, using resonance stabilisation of the conjugate bases.

Q.24 Distinguish between benzaldehyde and acetophenone using the iodoform test. Write the relevant reaction and observation.

Section D — Solutions and Reaction Kinetics

Q.25 A mixture of two liquids shows a positive deviation from Raoult's law. Explain the molecular reason for the deviation and state what happens to the temperature when the process is accompanied by positive enthalpy of mixing.

Q.26 A reaction follows

$$\text{Rate} = k[A]^m[B]^n.$$

Using the experimental observations provided in the question, determine m , n and the rate constant.

The original numerical tables and several MCQ stems are not fully recoverable from the scanned answer booklet.